Evaluation of feature selection methods

Data exploration and visualization - exercise 1

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Goal

Evaluate following feature selection methods:

- Lasso (embedded)
- Random Forest Importance score (embedded)
- Mutual Information (filter)
- Correlation (filter)
- BIC - Bayesian Information Criterion (wrapper)

and dimensionality reduction techniques:

- PCA - Principal Component Analysis
- TruncatedSVD
- Factor Analysis
- Isomap

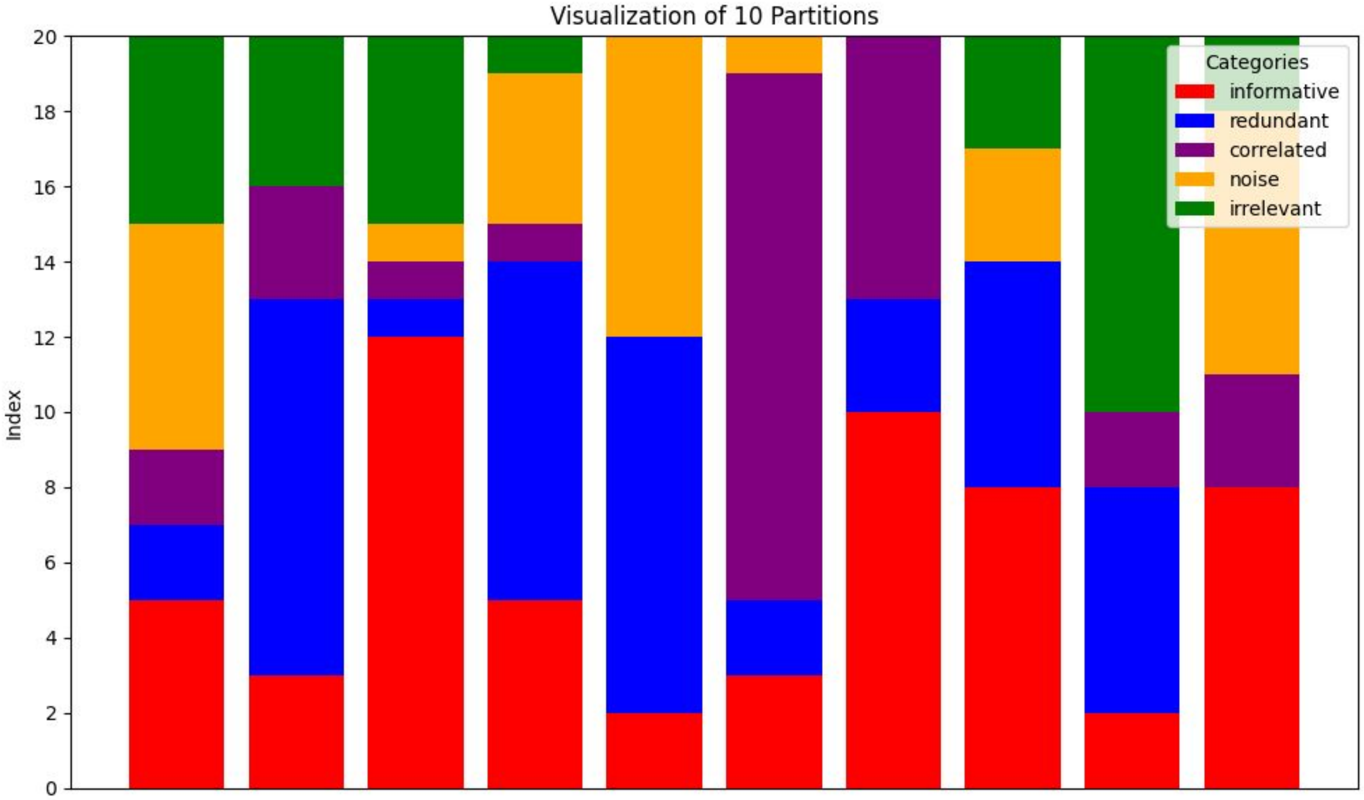
Pipeline

1. Set Seed
2. Repeat n times:
    2.1. generate column type partitions
    2.2. generate data according with column type configuration using `make_classification`
    2.3. split data on train and test dataset
    2.4. run each feature selection and dimensionality reduction method
    2.5. for each new representation of data:
        2.5.1. train SVM and Random Forest classifiers on train set
        2.5.2. on test set calculate metrics: accuracy, precision, recall, F1
3. Aggregate results
4. Make visualizations

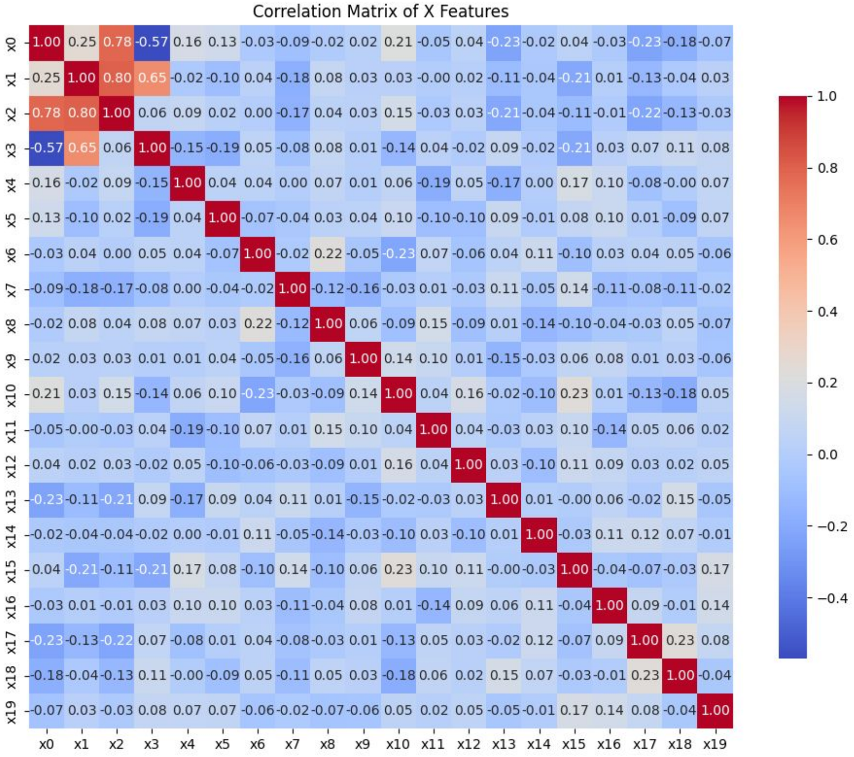
Data Generation - column type

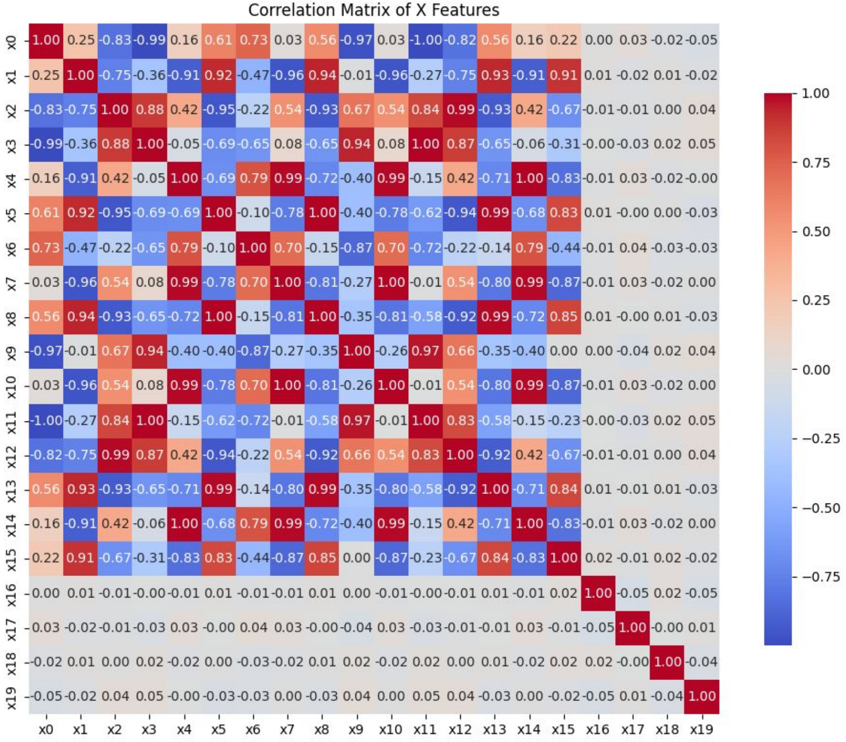
Informative	Generated by make_classification(), directly contribute to determining the class label.
Redundant	Generated by make_classification(), random linear combinations of the informative features.
Correlated	Copies of informative and redundant features with vector of N(0,0.2) added.
Very noisy	Copies of informative and redundant features with vector of U(-2,2) added.
Irrelevant	Generated by make_classification(), don't have any relation with class label.

Example dataset column partitions



Dataset Generation



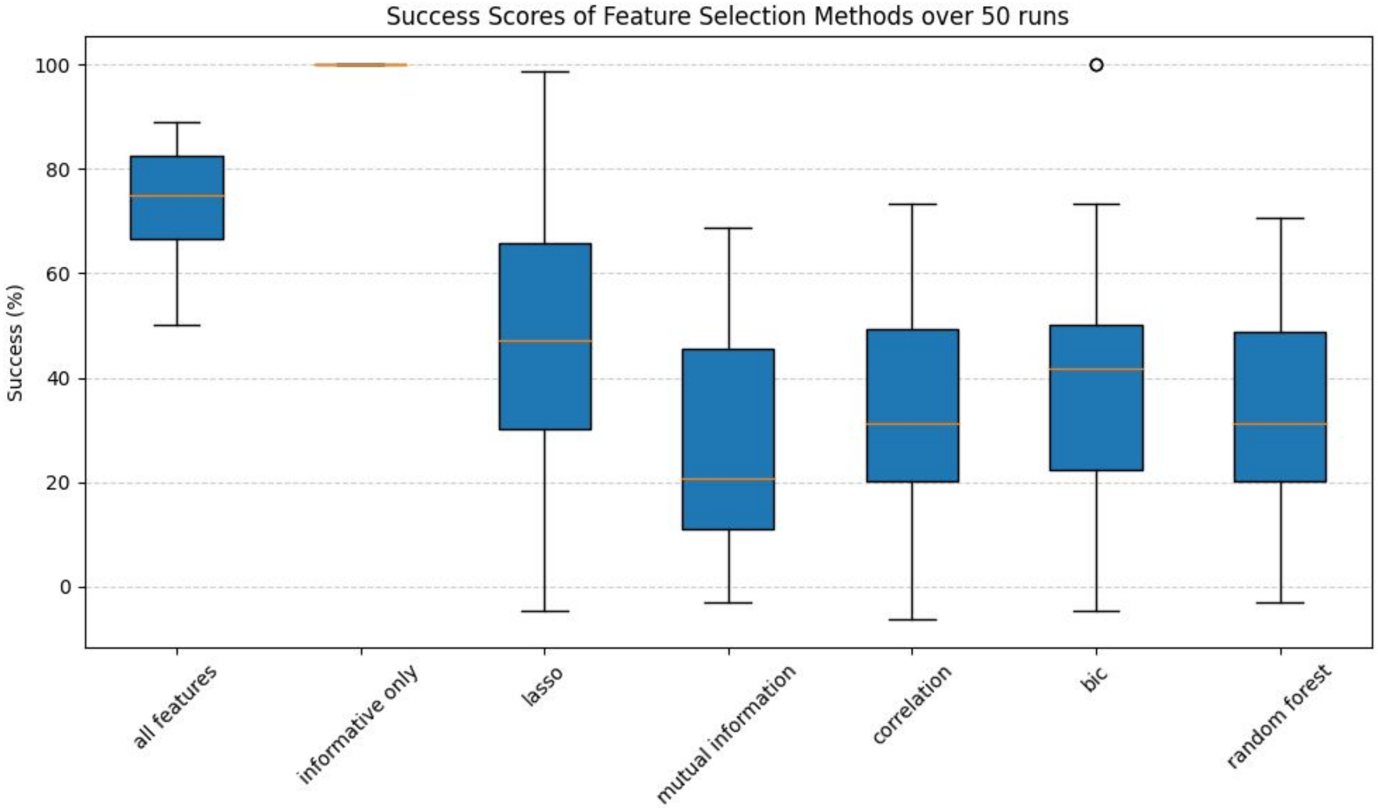


Example feature selection results - 1

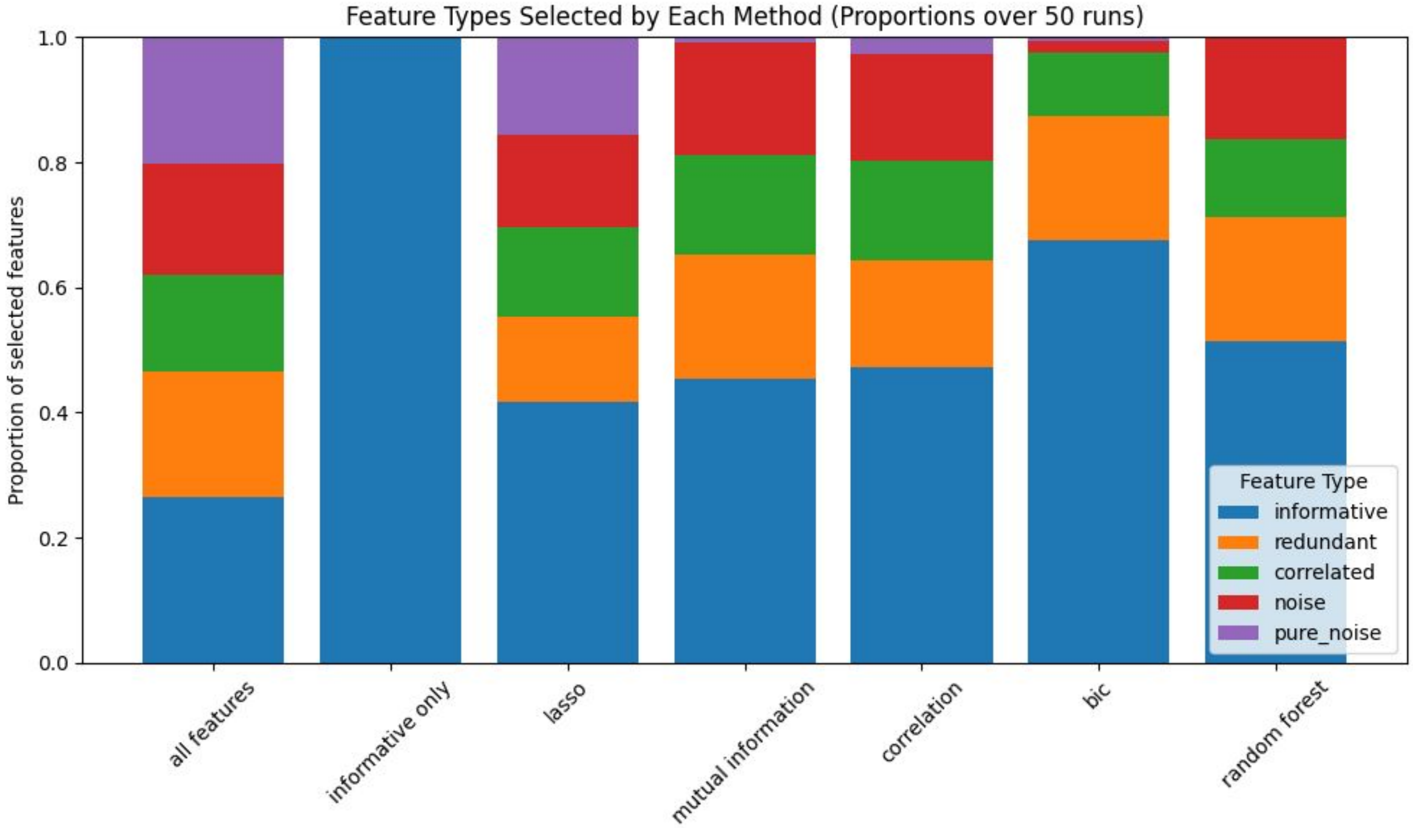
Example feature selection results - 2

Example feature selection results - 3

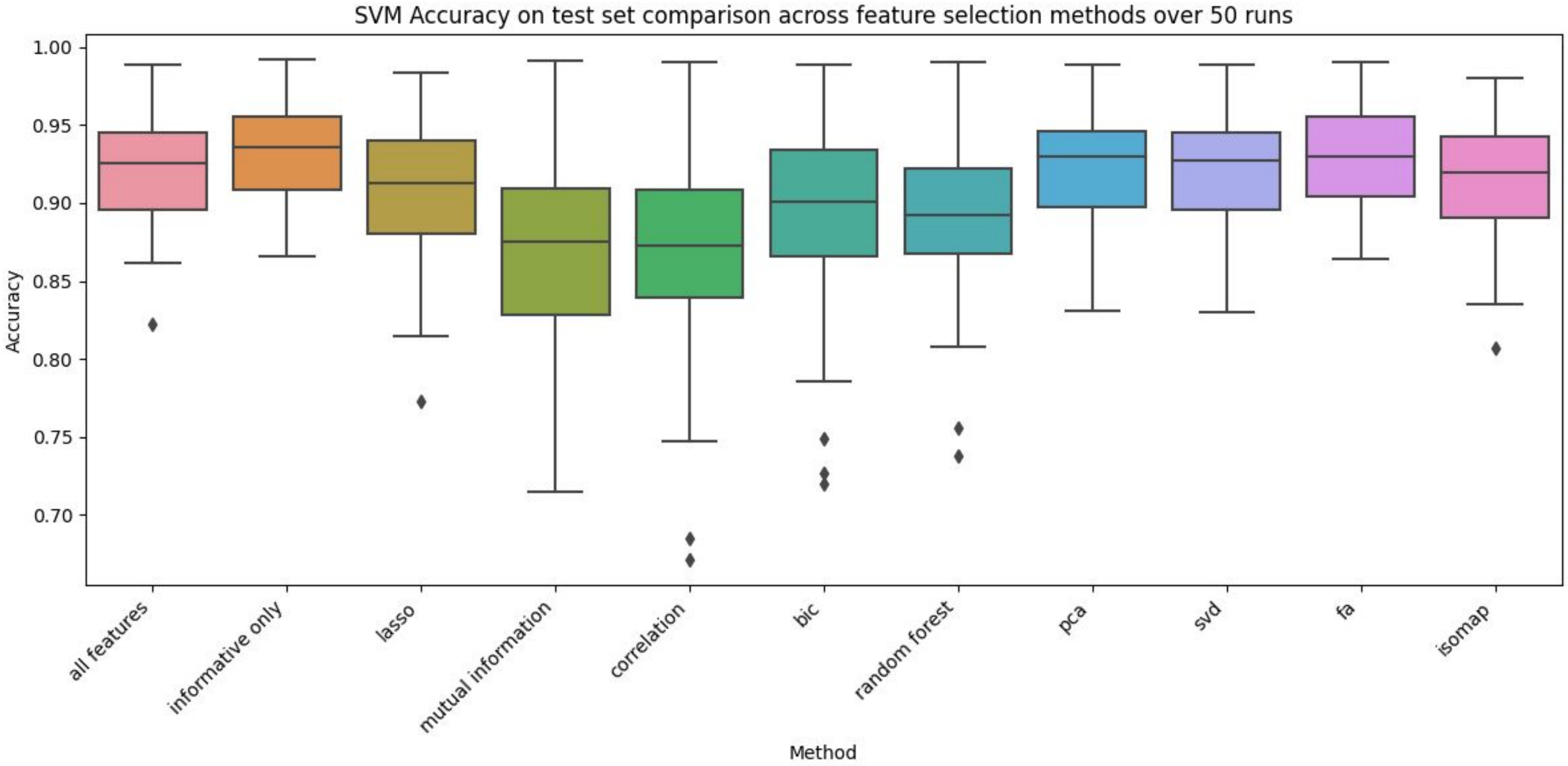
Success metric as a evaluation metric



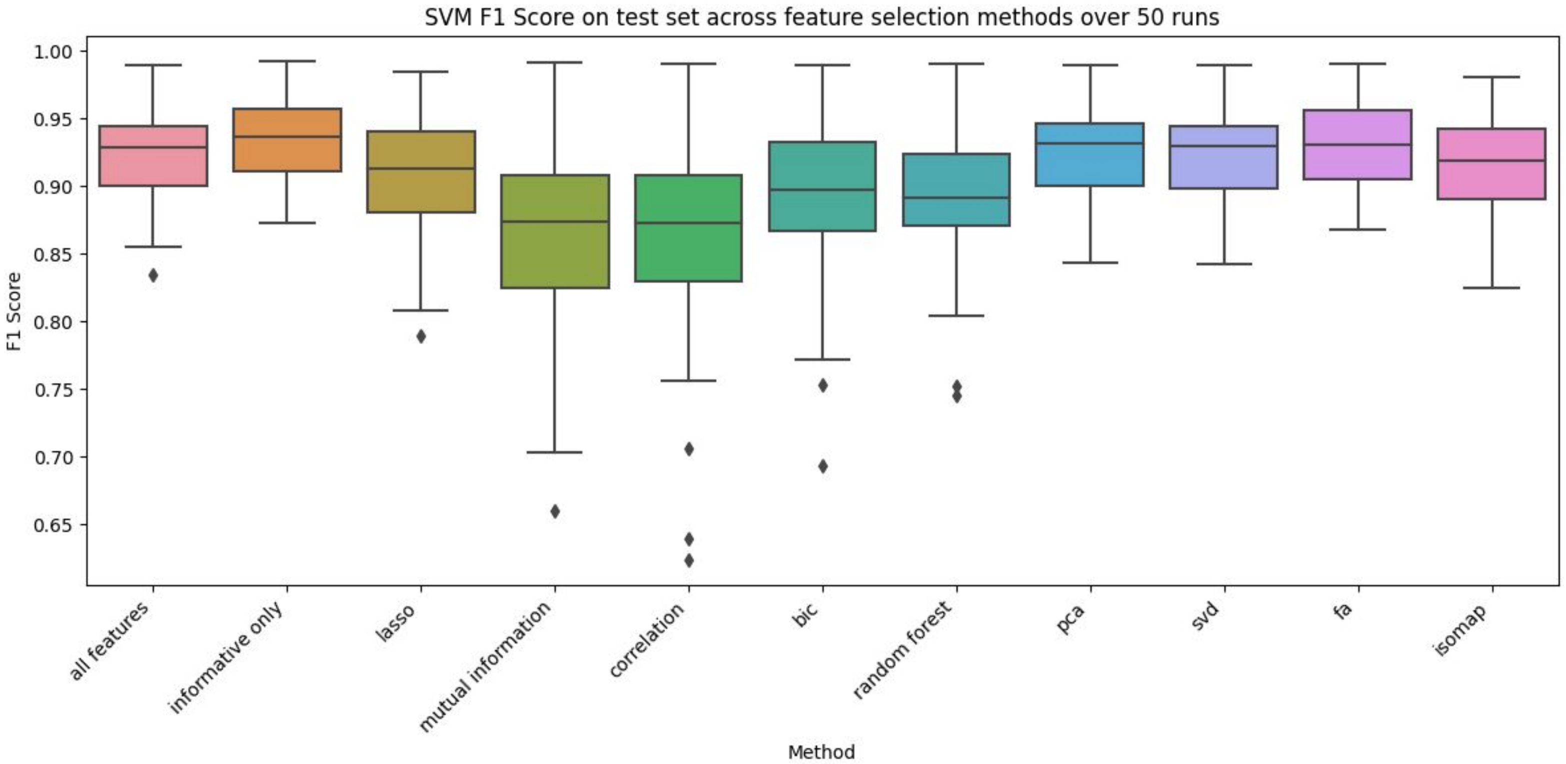
Other method of visualization the selection performance



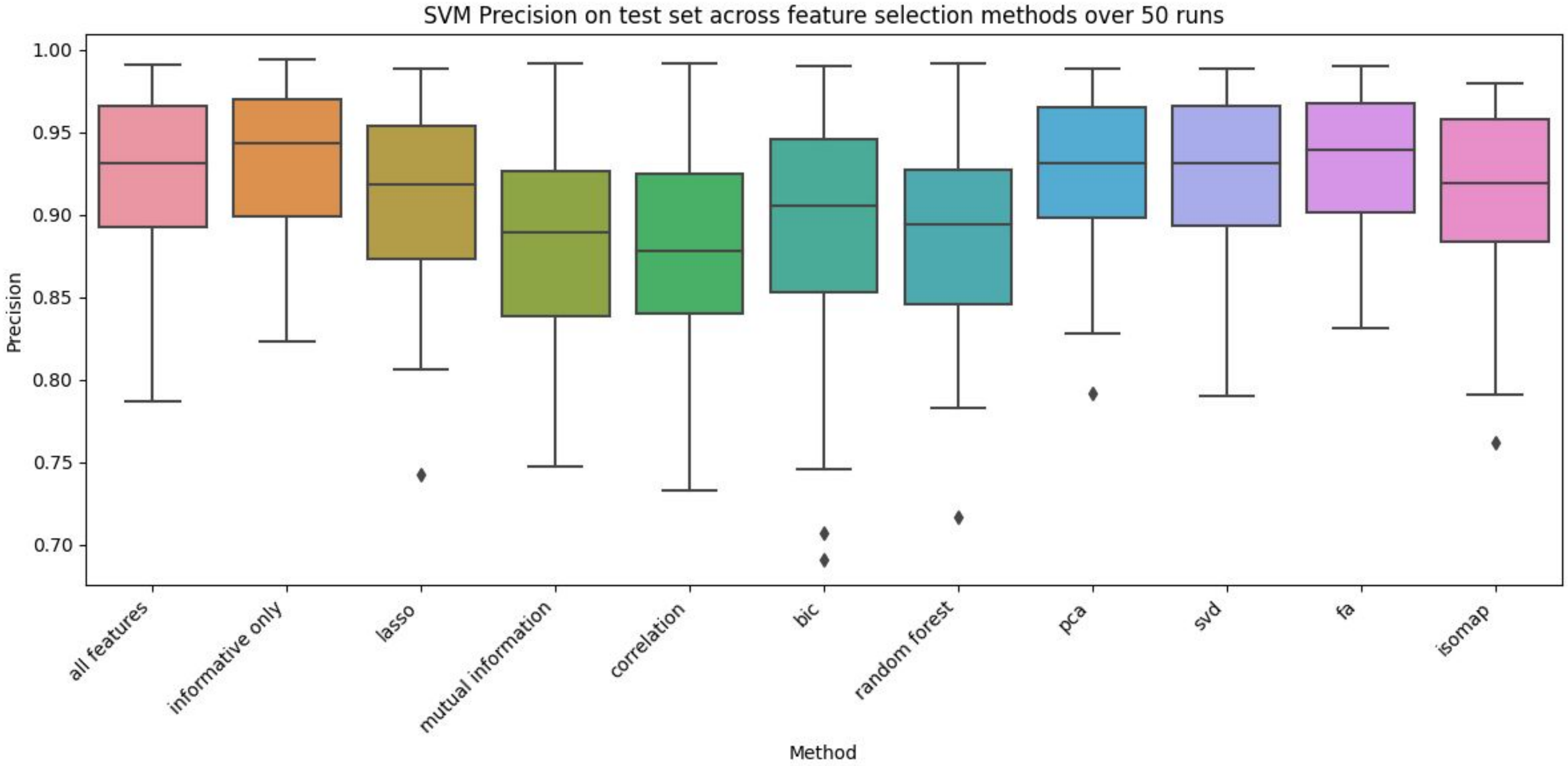
Evaluation of classification results - 1



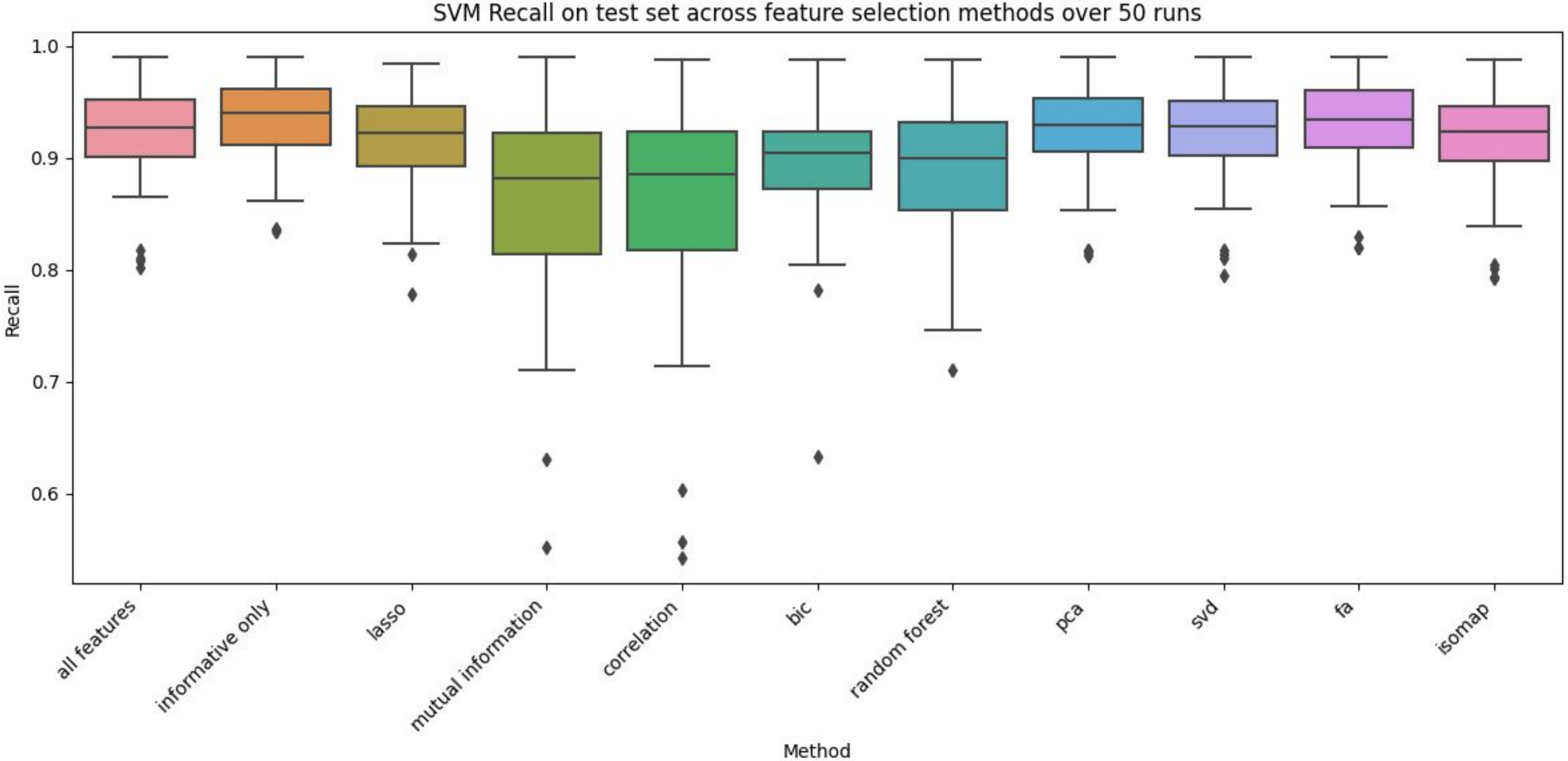
Evaluation of classification results - 2



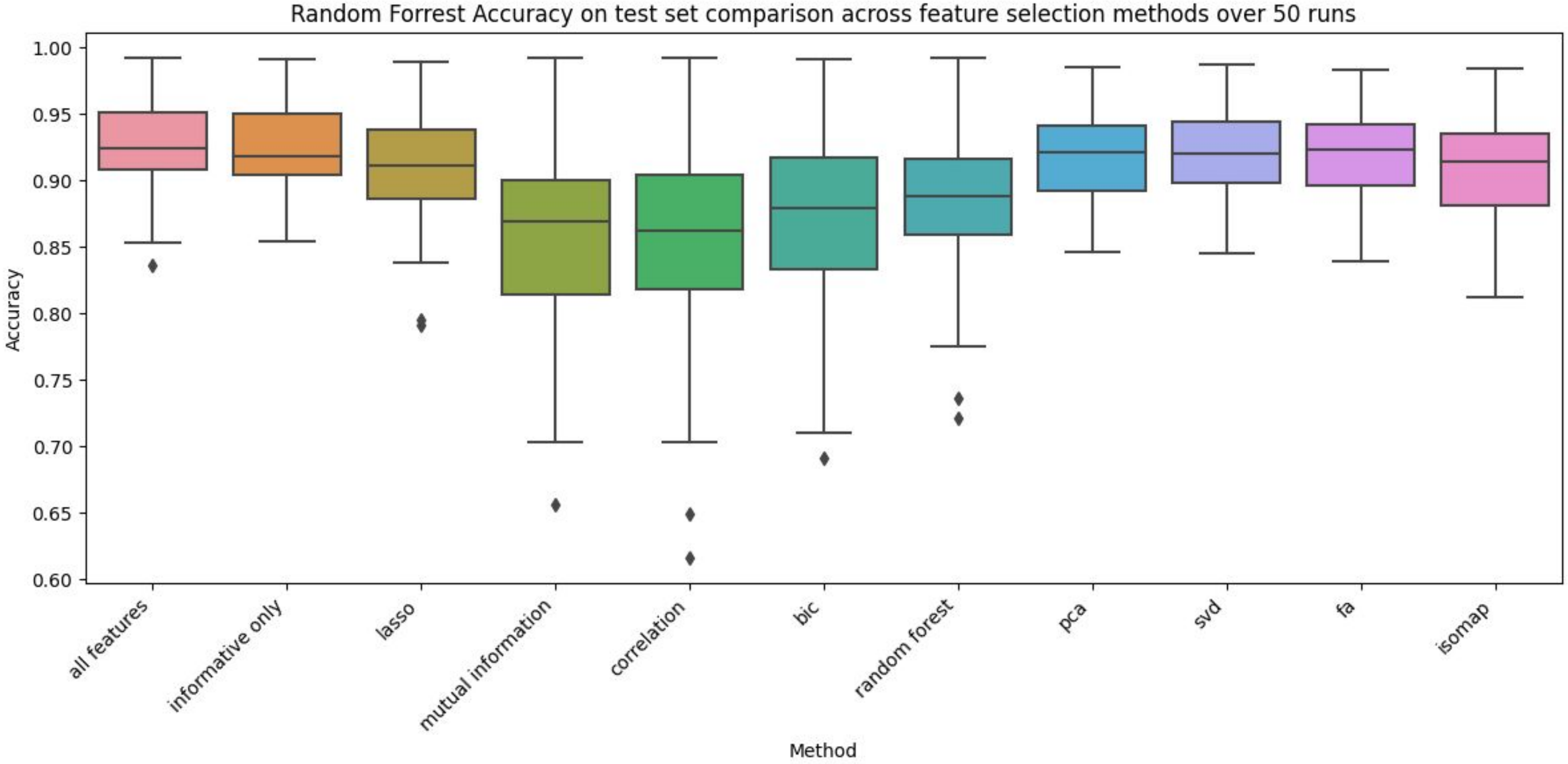
Evaluation of classification results - 3



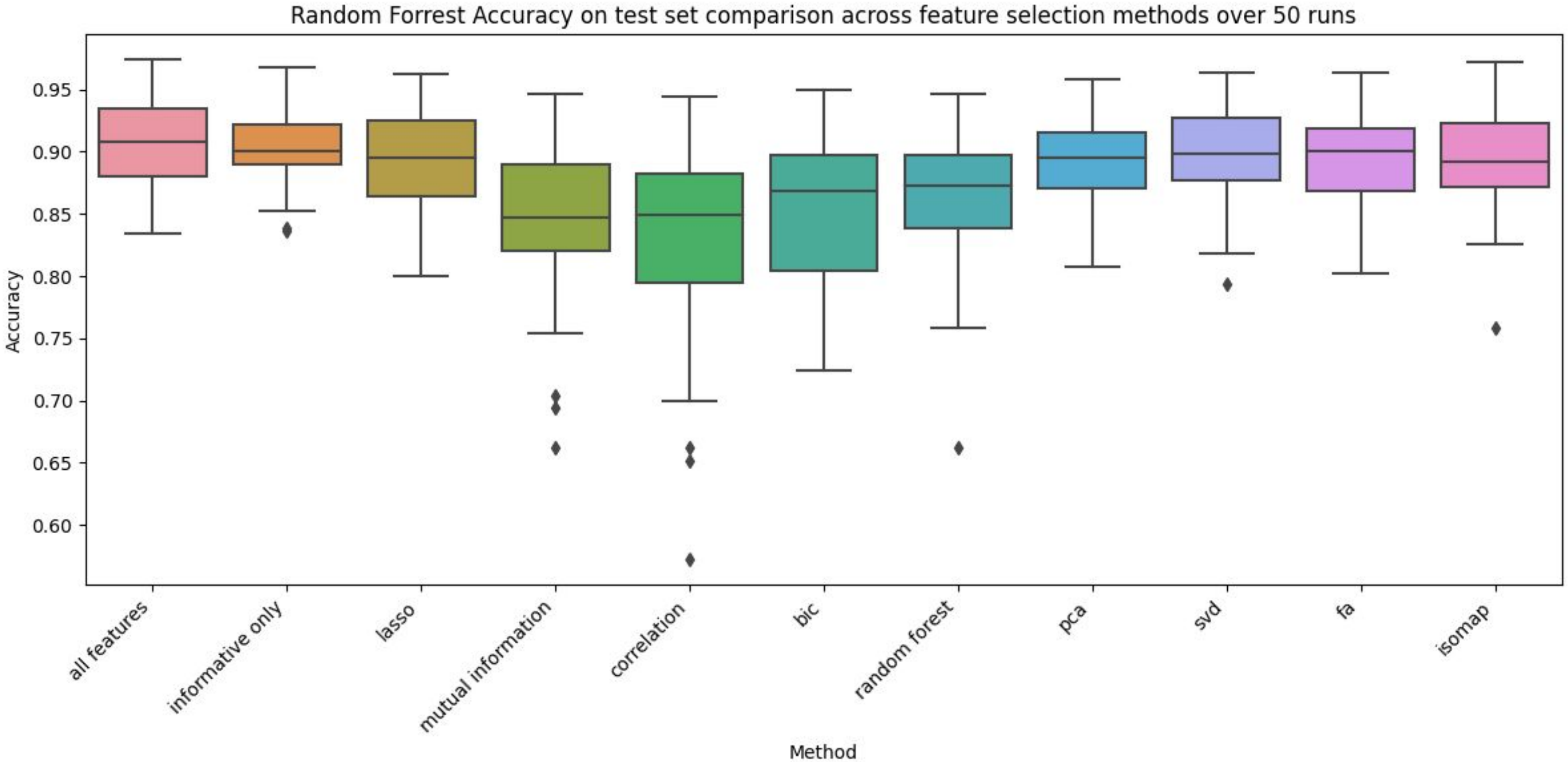
Evaluation of classification results - 4



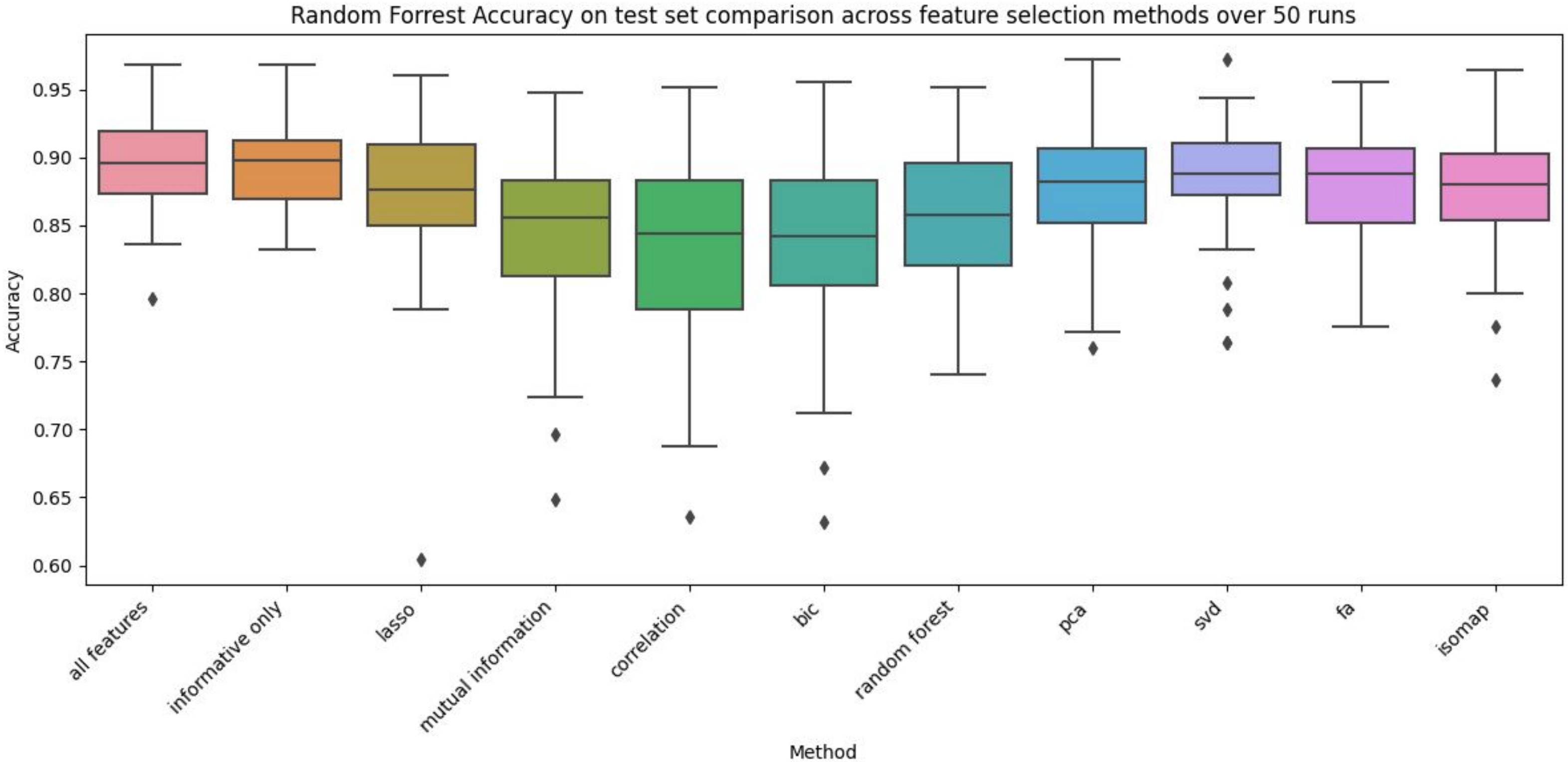
Classification results and set size = 2000



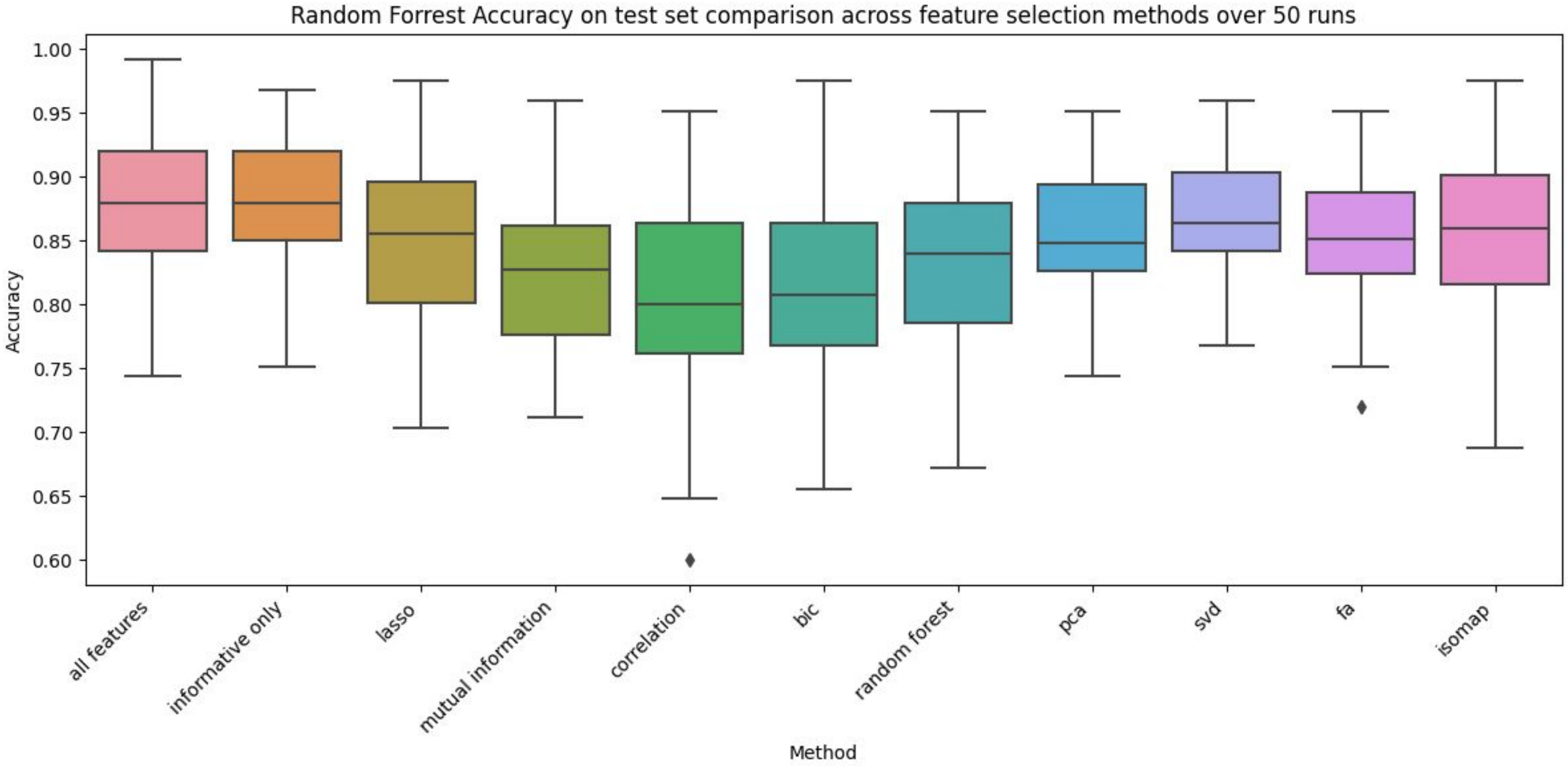
Classification results and set size = 1000



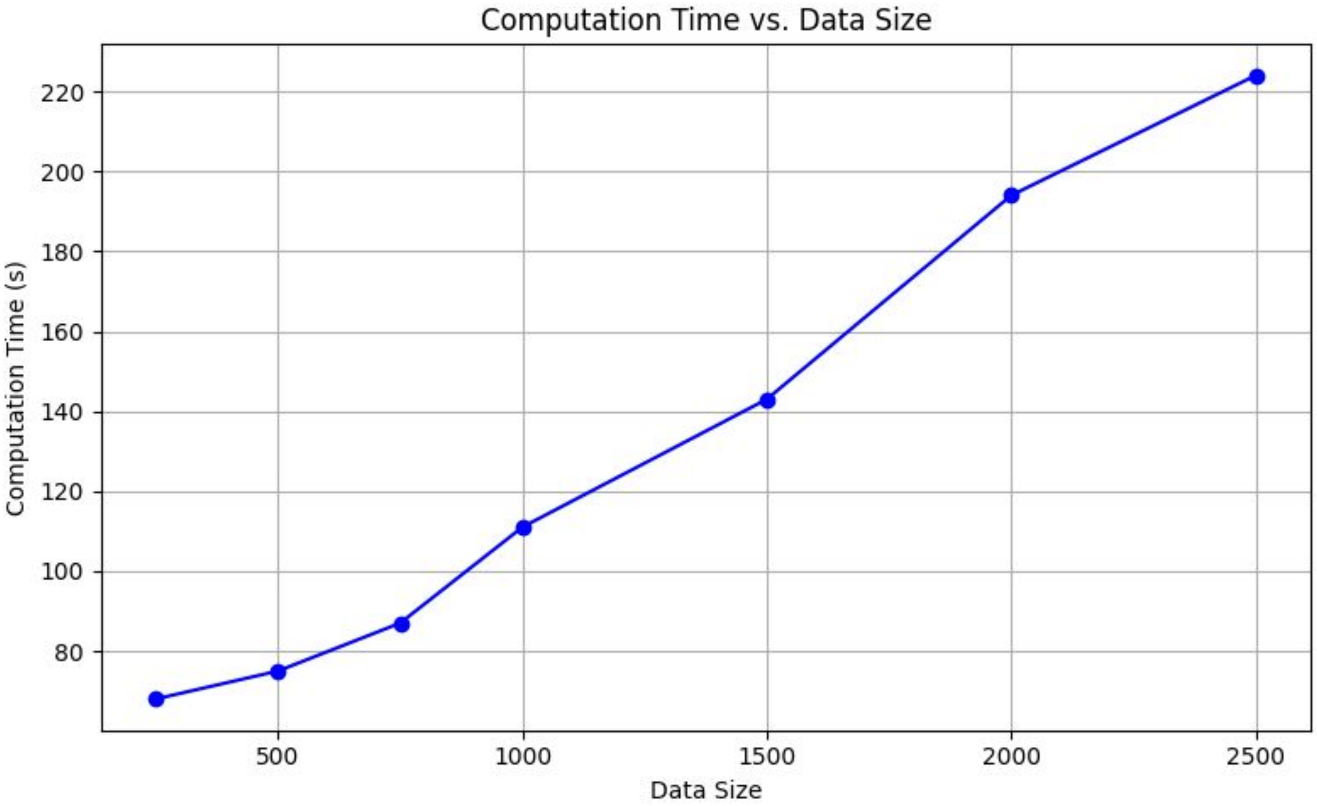
Classification results and set size = 500



Classification results and set size = 250



Computational complexity estimation



Evaluation of clustering algorithms

Data exploration and visualization - exercise 2

Mateusz Nizwantowski

Goal

Evaluation of data clustering algorithms:

- K-means
- Genie ($g \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$)
- Agglomerative Hierarchical Clustering (with single, average, complete and Ward linkage)
- DBSCAN (eps = 0.2, minimum number of samples = 5)

Pipeline

1. Initialize & Configure

2. For each Dataset (Load X, y_trues, k_values):

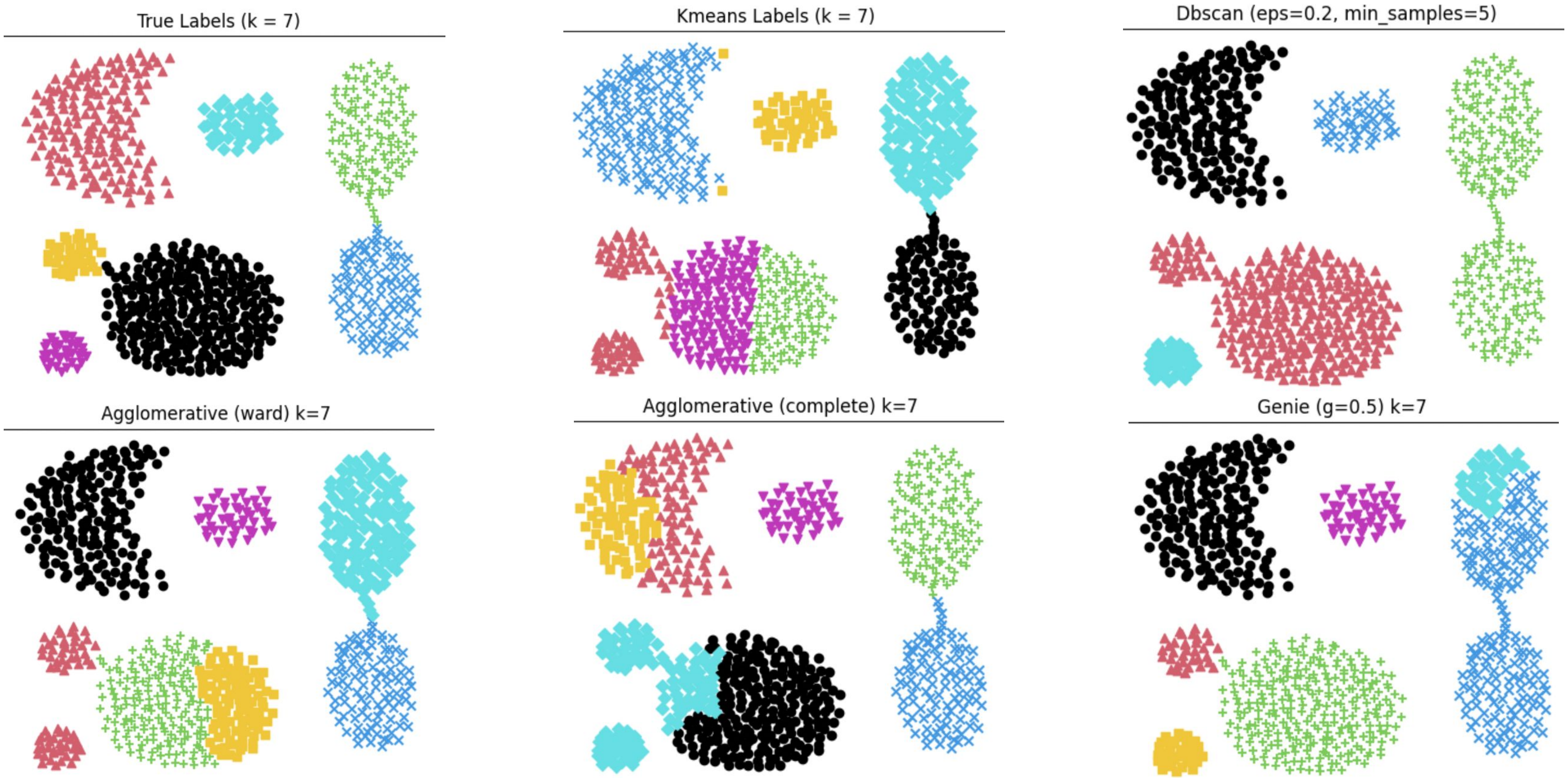
For each k in k_values run:

- **KMeans**
- **For each gini threshold separate Genie**
- **For each linkage type separate agglomerative**
- **DBSCAN**

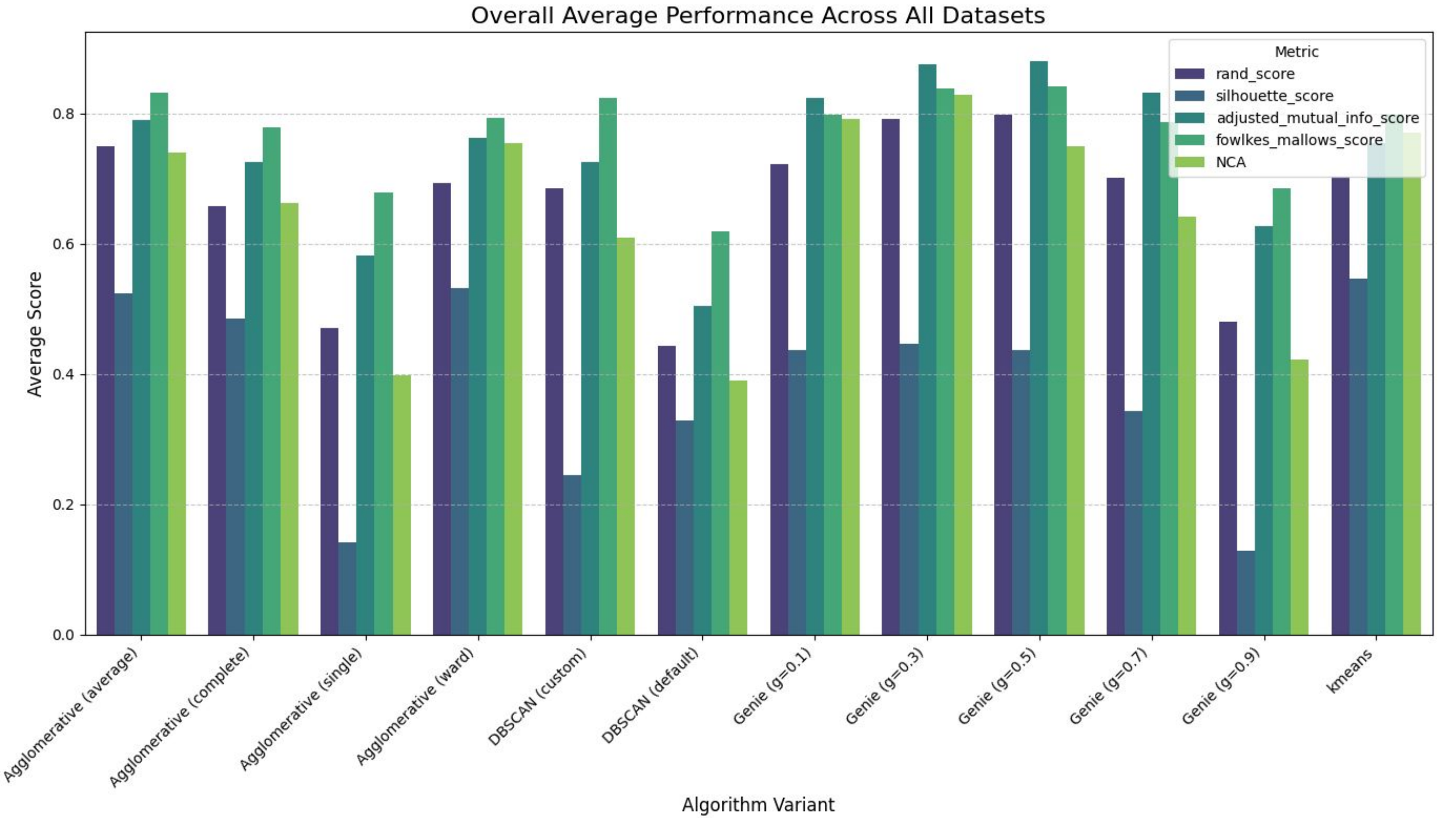
For each of those configuration - Fit, predict, evaluate all metrics, and store results

3. Save the complete results DataFrame to a single CSV file to use it for further analysis.

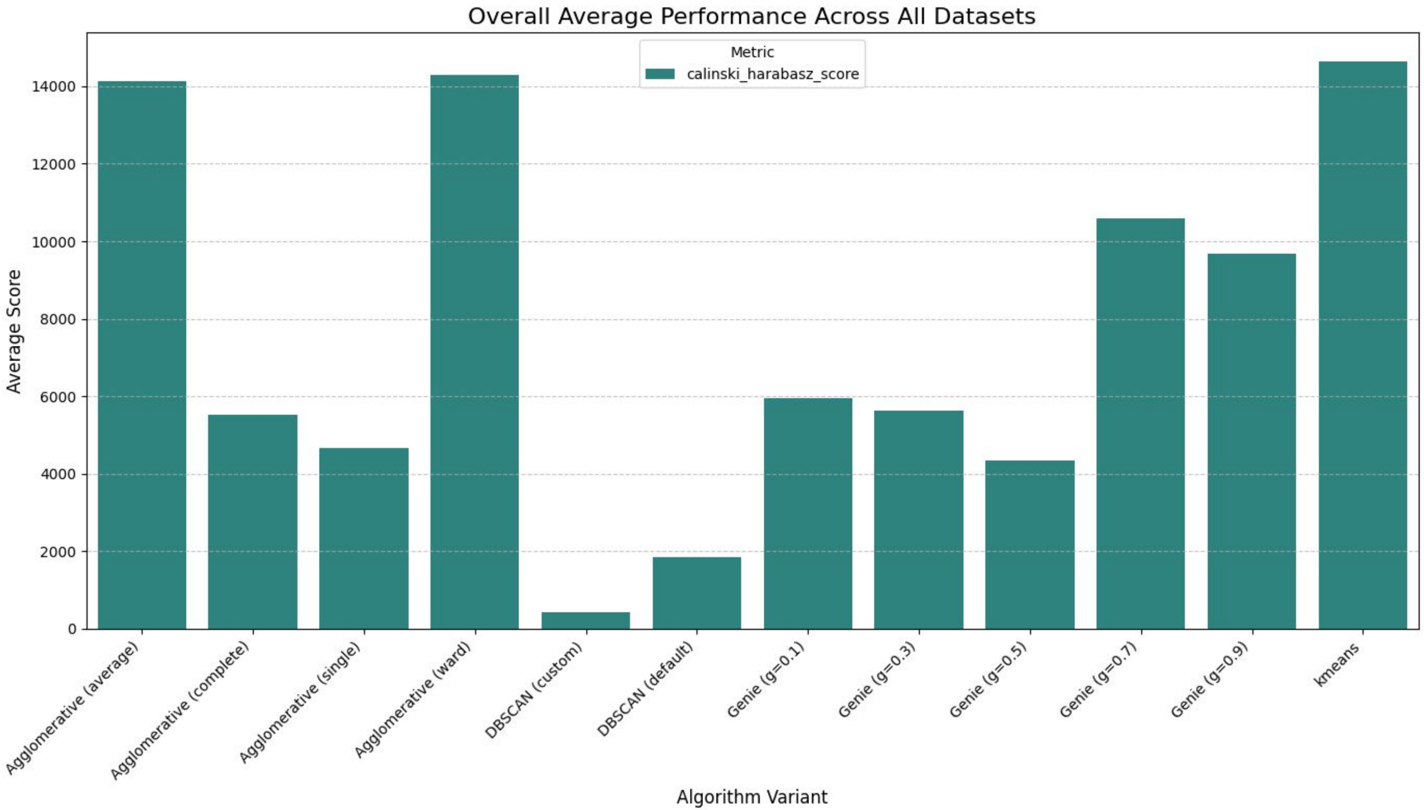
Clustering visualization



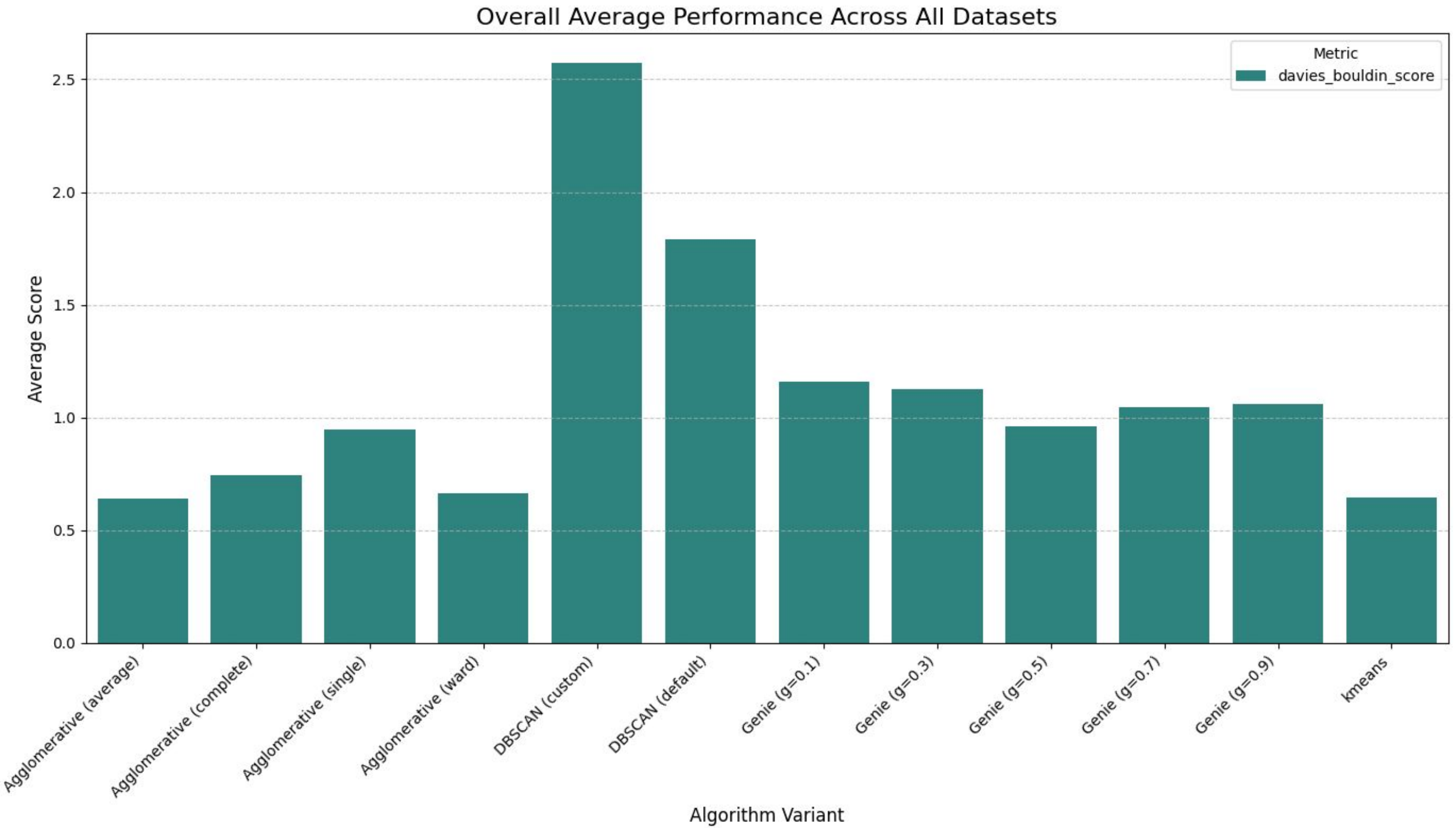
Average performance for each configuration - 1



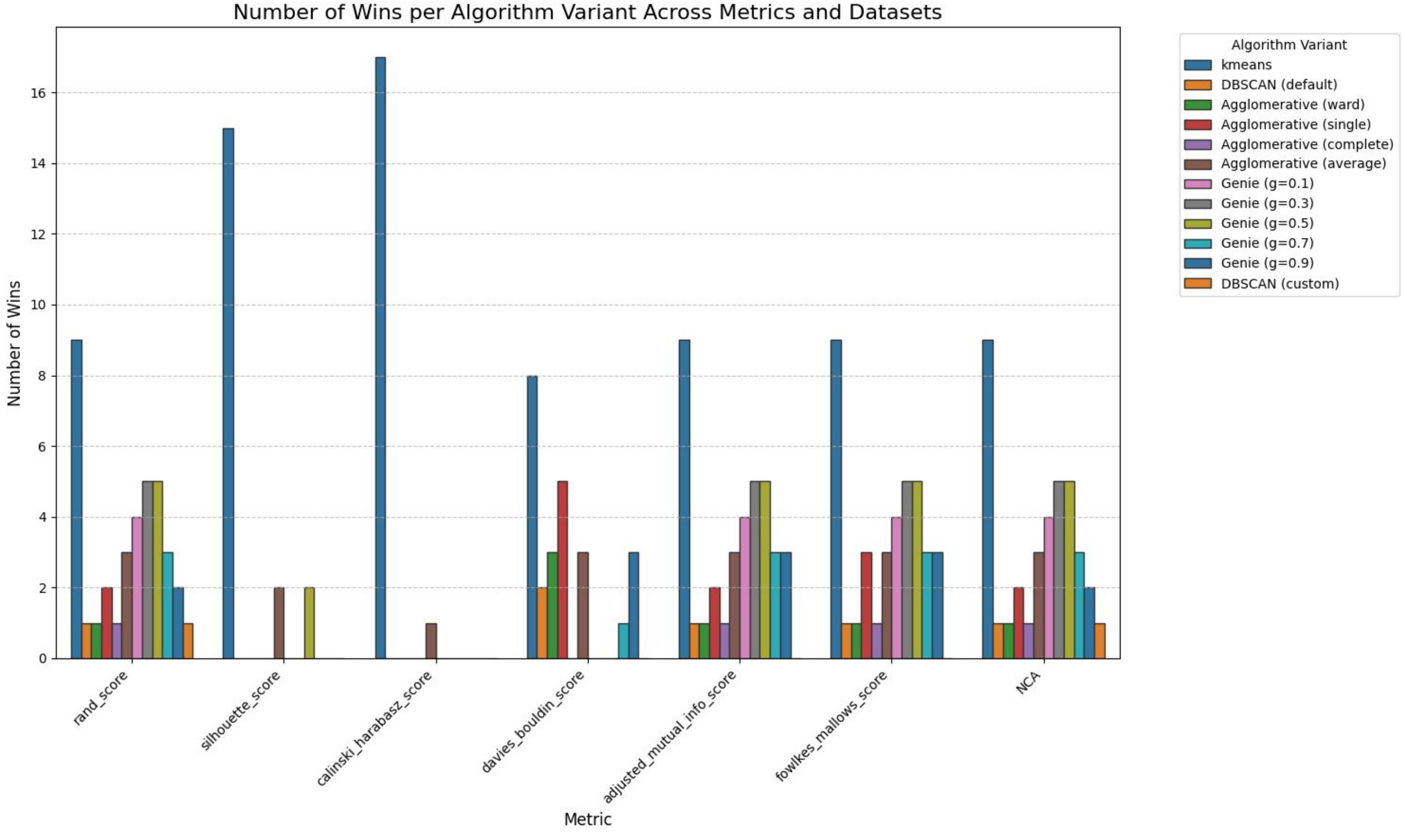
Average performance for each configuration - 2



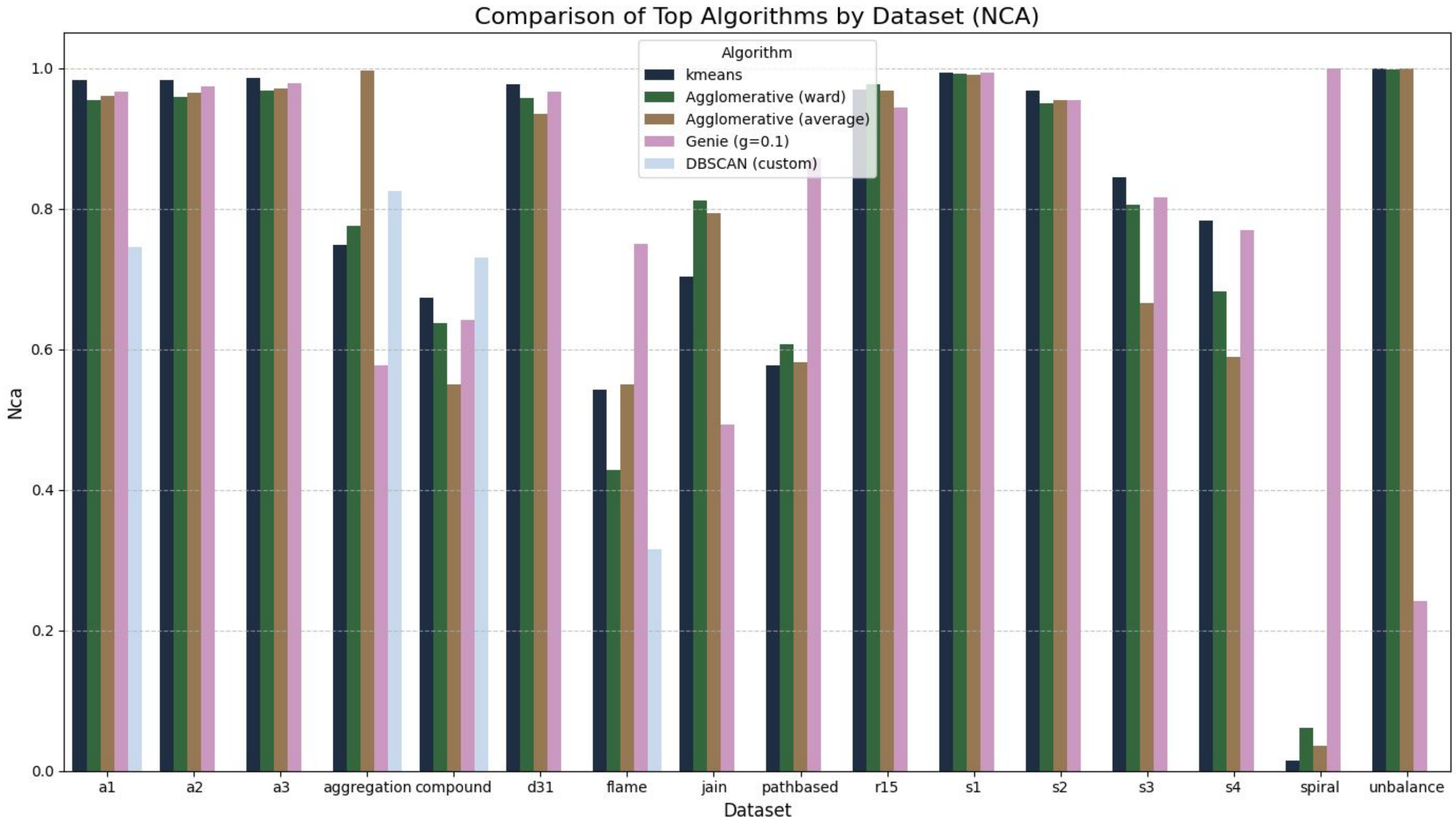
Average performance for each configuration - 3



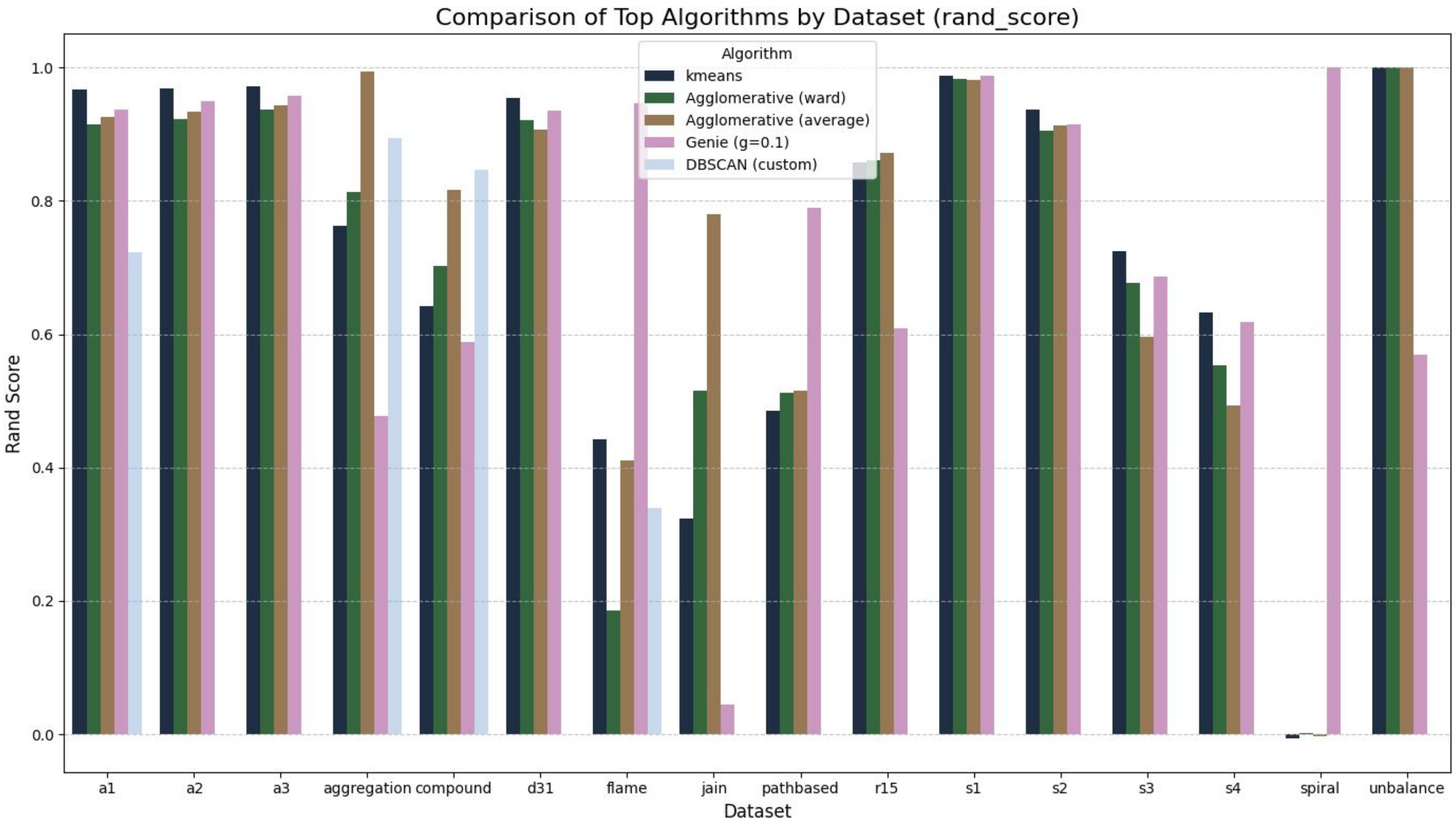
Summary of performance



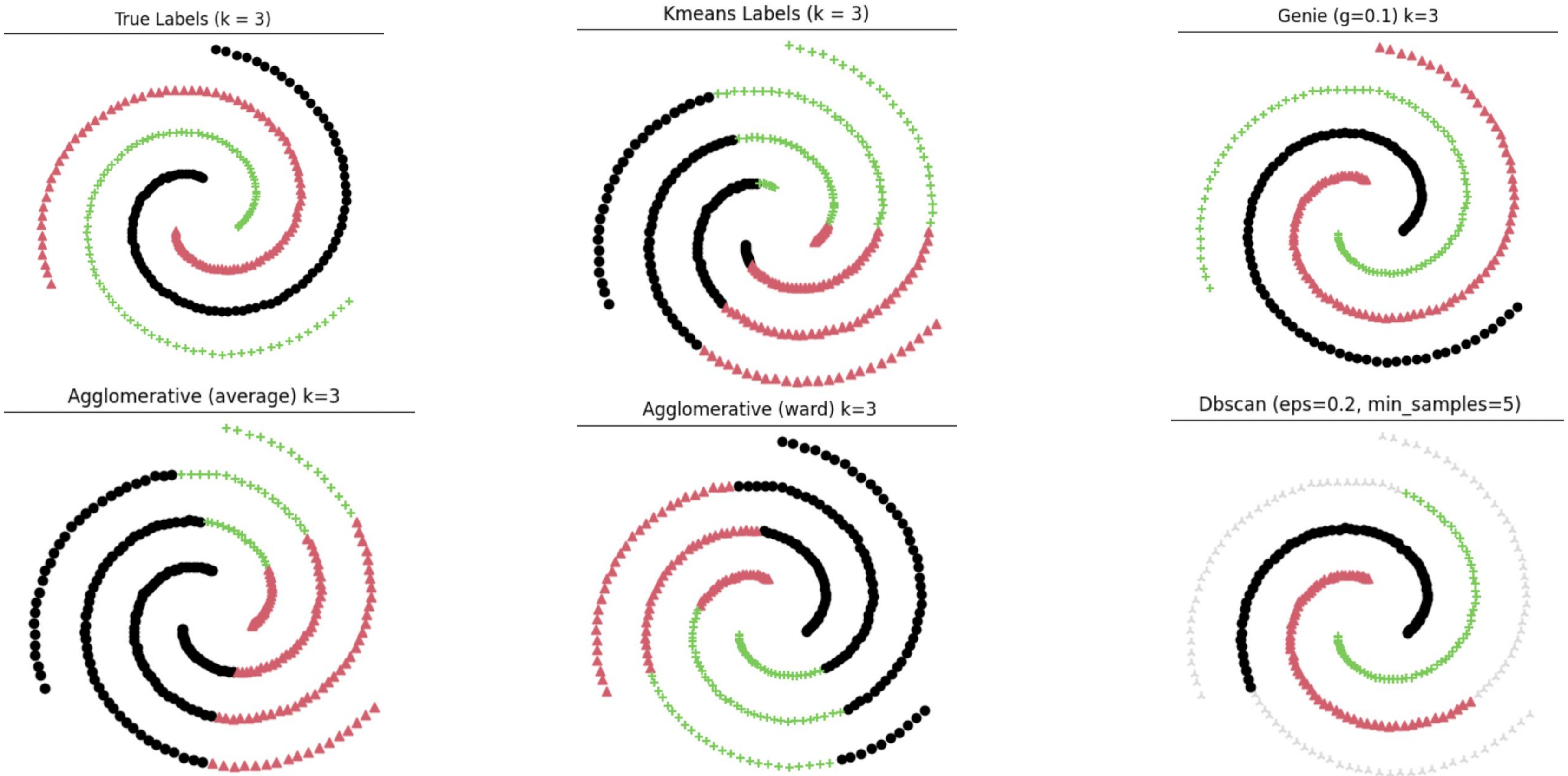
Sanity check of results - 1



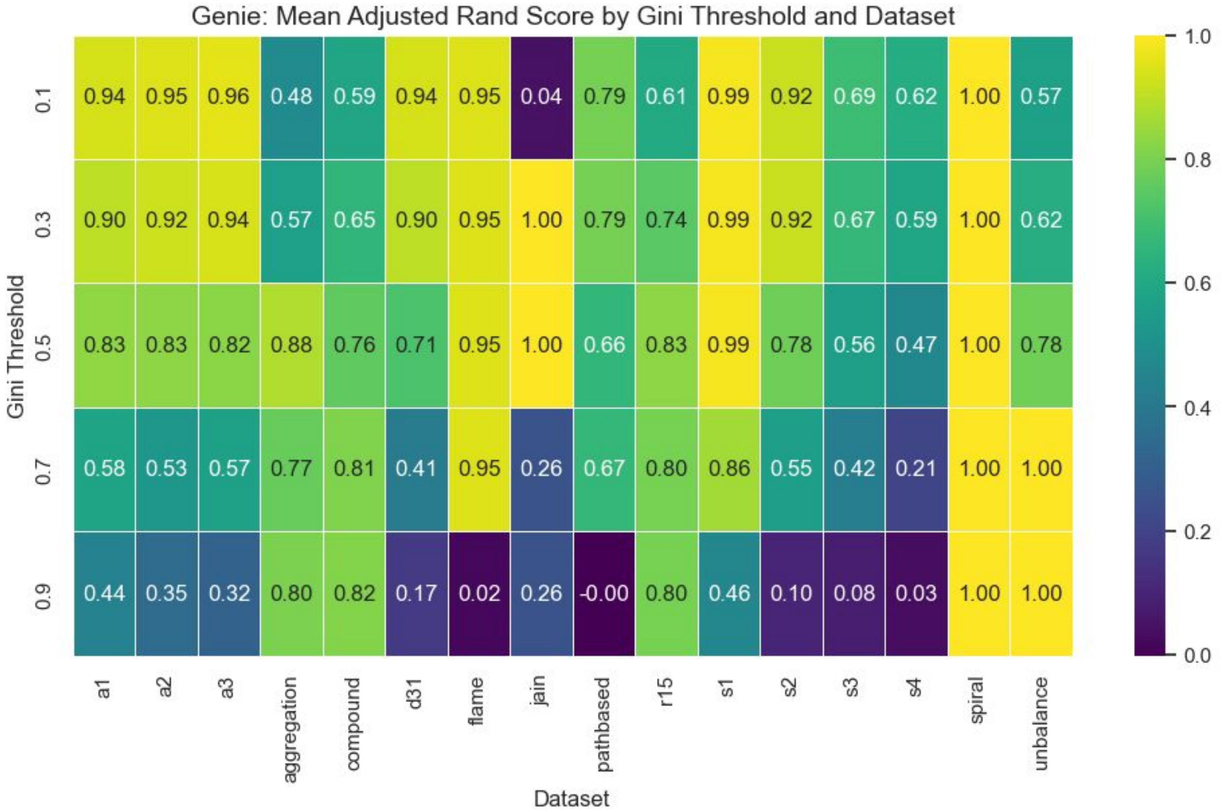
Sanity check of results - 2



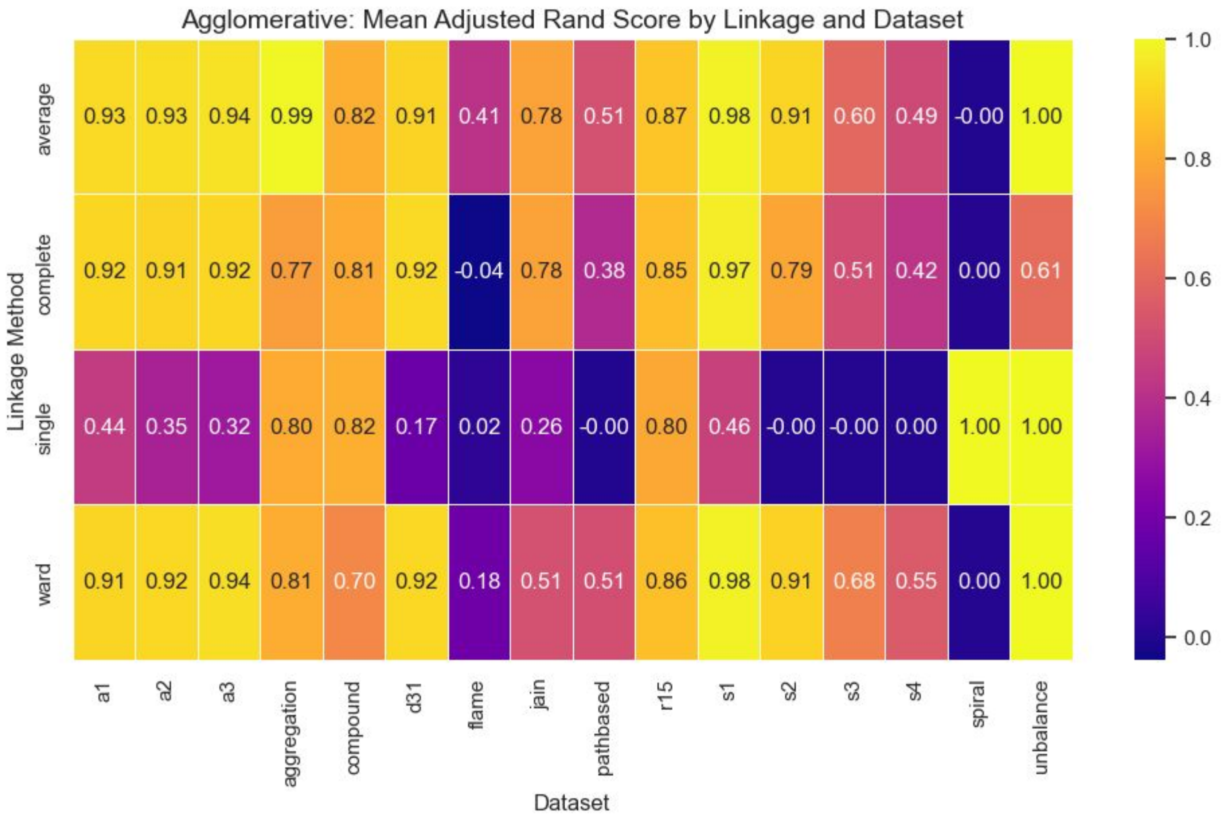
Clustering visualization - Spiral



Adjusted Rand Score for different thresholds



Adjusted Rand Score for different Linkage Methods



Thank you for your attention

Are there any questions?